IMAGE BASED ANIMAL TYPE CLASSIFICATION FOR CATTLE AND BUFFALO

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PROBLEM STATEMENT

IMAGE BASED ANIMAL TYPE CLASSIFICATION FOR CATTLE AND BUFFALO :

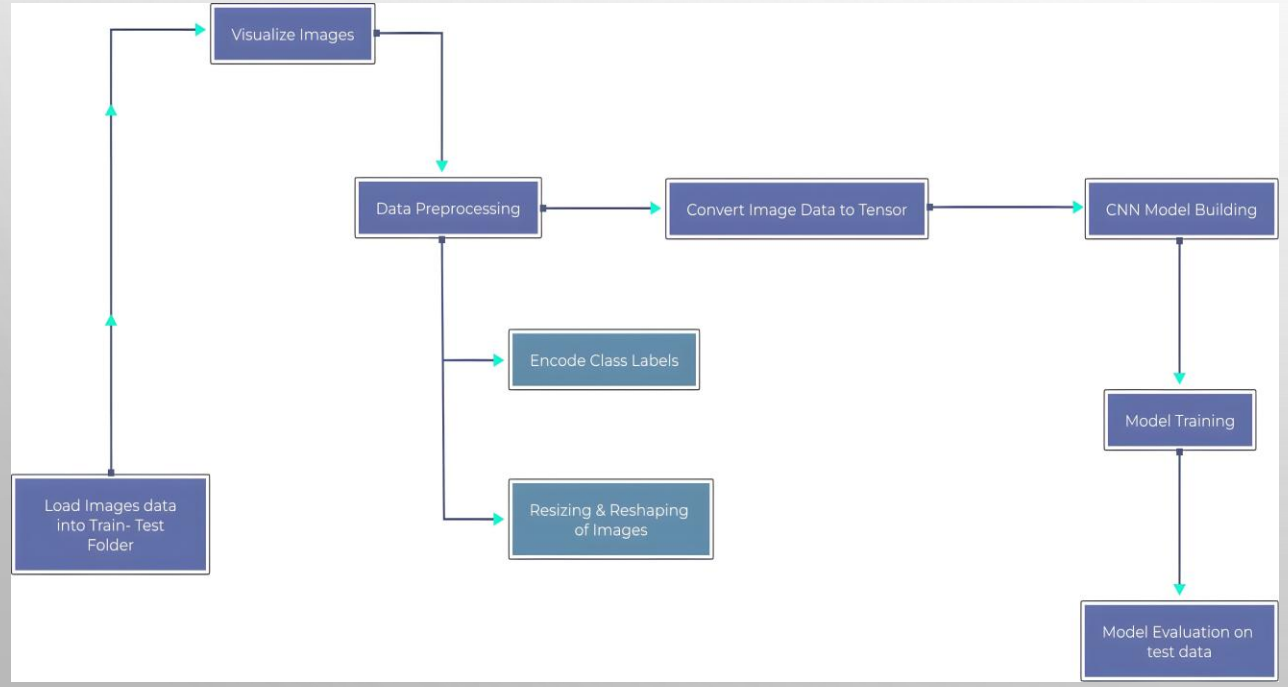
- Cows and buffaloes look very similar in **color, size, and body structure**, making it difficult to distinguish them from images.
- Identifying animals manually can lead to mistakes, especially when done quickly or without proper attention.
- In farms or **large datasets**, checking each image manually takes **a lot of time and effort**.
- There are **very few simple and user-friendly systems** available to automatically classify animals using images.

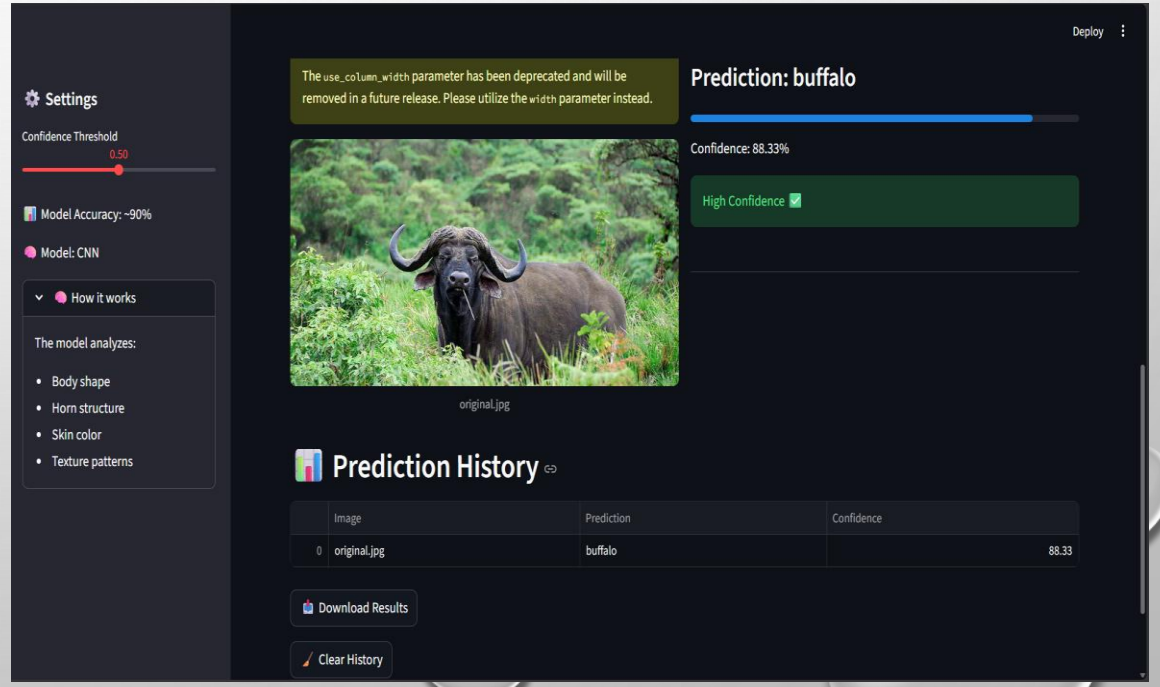
PROPOSED SOLUTION

- Develop a **machine learning model** to classify animals
- Use **convolutional neural network (CNN)** for image recognition
- Train the model with images of **cows and buffaloes**
- Upload image → system predicts whether it is **cow or buffalo**
- Display result with **confidence percentage**
- Provide a simple **user-friendly interface (UI)** for easy usage

TECHNICAL APPROACH

- **Python** – Used to write the main program
- **TensorFlow/Keras** – Used to build and train the CNN model
- **PIL / OpenCV** – Used for image processing (resize, convert)
- **Streamlit** – Used to create simple user interface





FEASABILITY AND VIABILITY

Feasability	Viability
Easy to develop using existing ML tools (Python, TensorFlow)	Useful in real-world (farming, dairy sector)
Requires only basic system and dataset	Saves time and reduces manual effort
Can be implemented in short time	Can be expanded to detect more animals
Simple UI makes it easy to use	Cost-effective and scalable solution

RESEARCH AND REFERENCE

- Image classification using convolutional neural networks — IEEE
  https://ieeexplore.ieee.org/document/726791
- Deep learning for image recognition — google research
  https://research.Google/pubs/pub43146/
- convolutional neural networks for visual recognition — stanford university
  http://cs231n.stanford.edu/
- tensorflow image classification tutorial — google
  https://www.Tensorflow.Org/tutorials/images/classification
- Animal image classification using CNN — kaggle
  https://www.Kaggle.Com/code